

Project Description Document

Group Members

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Group Name

Free Labour

The Problem

Description

This project seeks to model the process by which human evaluators determine whether a candidate is qualified for a job and to implement a system that can approximate this decision process using computational methods.

The system will analyse candidate resumes and job descriptions, extract relevant competencies from both, and determine whether meaningful matches exist between them. Candidate resumes and job postings will be treated as primary inputs, and natural language processing techniques will be used to extract structured representations of skills, experience, education, and other relevant competencies from unstructured text.

For each job, the system will identify candidates whose competency profiles satisfy the required criteria extracted from the job description and present them in ranked order of overall suitability using configurable weighting of relevant criteria. For each candidate, the system identifies jobs whose required criteria are satisfied by the candidate's competency profile and presents them in ranked order of compatibility and relevance. Where few or no suitable matches exist, the system will provide feedback indicating gaps between candidate competencies and available job requirements.

A candidate will be considered qualified for a job when the candidate's extracted competency profile satisfies the required criteria derived from the job description. Qualification will be

determined across multiple dimensions rather than by a single numerical threshold or score. These dimensions may include required skill coverage, relevance of experience, educational compatibility, and semantic similarity between candidate competencies and job requirements. Once required criteria are satisfied, candidates may be ranked according to their overall suitability for a given role, and jobs may be ranked according to their relevance to a given candidate.

The objective is to minimise incorrect recommendations and missed valid matches by approximating human judgement of job suitability while improving transparency in how matches are determined. An incorrect recommendation occurs when a candidate is identified as suitable despite lacking required competencies, while a missed match occurs when a candidate who satisfies required competencies is not identified. These conditions will guide evaluation of system performance.

This objective is stated under the assumption of accurate extraction of competencies from CV and job description text. The system also assumes that resumes and job descriptions reasonably reflect actual competencies and requirements, and that competency extraction techniques provide sufficiently accurate representations for comparison. The scope is limited to computing-related job roles to ensure manageable dataset size and realistic evaluation.

It should be noted that the system functions as an initial matching and recommendation stage rather than a complete hiring solution, and is designed to operate within feasible computational limits for the dataset sizes considered.

Motivation

Modern recruitment systems often fail to transparently and efficiently match qualified candidates to appropriate job opportunities. Applicants frequently submit resumes without understanding why they are rejected and without receiving meaningful feedback, while employers must manually filter large numbers of applications with varying relevance. This misalignment leads to missed opportunities for qualified candidates and inefficient hiring processes for employers.

System Functionalities

For Both Users:

- Determine whether candidate competencies satisfy required job criteria
- Ensure only candidates satisfying required competencies are recommended for a job
- Maintain transparency by showing which competencies contributed to a match decision
- Operate as a prototype matching and recommendation stage rather than a full hiring platform

For Employers:

- Upload and parse job descriptions
- Extract required and preferred competencies from job descriptions
- Represent job data as structured competency profiles
- Rank qualified candidates for each job by overall suitability
- Provide explanations indicating why a candidate is or is not matched to a job
- Ensure only candidates satisfying required competencies are recommended for a job
- Provide configurable weighting of evaluation criteria for ranking purposes
- Display process status for each employer (e.g., description uploaded, matching in progress, matches identified, no suitable matches found, waiting on candidate response)

For Candidates:

- Upload and parse candidate CVs
- Extract candidate competencies from resumes
- Represent candidate data as structured competency profiles
- Rank relevant job opportunities for each candidate by compatibility
- Provide feedback to candidates when few or no suitable matches exist
- Ensure candidates are shown only jobs for which required competencies are satisfied
- Identify missing competencies preventing qualification
- Display process status for each candidate (e.g., CV uploaded, matching in progress, matches identified, no suitable matches found, waiting on employer response)

Possible Tools Used Towards The Solution

We have considered several tools and technologies to support the design and implementation of the proposed system. As it regards actual coding languages and frameworks we believe we shall use those listed below.

- Backend: Python-based RESTful API (Flask or FastAPI)
- Frontend: JavaScript, React
- Database: PostgreSQL
- Deployment: Docker, Render
- NLP / Entity Extraction: Python, spaCy (Named Entity Recognition (NER))
- Transformer-Based LMs: HuggingFace Transformers
- Data Collection: Kaggle datasets, publicly available computing job datasets
- Model Development: Jupyter Notebook, pandas
- Model Evaluation: Jupyter Notebook, pandas
- Potential Evaluation Metrics:
 - Precision@K,
 - Recall@K,
 - nDCG@K,
 - Mean Reciprocal Rank

Names of Desired Supervisors

Dr. Daniel Coore	Confirmed
Dr. Anderson	Requesting
Dr. Coy	Requesting